

SQL DML Commands

03 · Data Manipulation Language — Complete Reference Guide

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1. INSERT — Adding Data to Tables

Method 1: Manual INSERT using VALUES

→ *Insert new records into the customers table*

```
INSERT INTO customers (id, first_name, country, score)
VALUES
    (6, 'Anna', 'USA', NULL),
    (7, 'Sam', NULL, 100)
```

→ *Insert a new record with full column values*

```
INSERT INTO customers (id, first_name, country, score)
VALUES (8, 'Max', 'USA', 368)
```

→ *Insert without specifying column names (not recommended)*

```
INSERT INTO customers
VALUES (9, 'Andreas', 'Germany', NULL)
```

→ *Insert with only id and first_name (other columns will be NULL or default)*

```
INSERT INTO customers (id, first_name)
VALUES (10, 'Sahra')
```

■ Common Errors to Avoid:

- Incorrect column order in VALUES
- Incorrect data type in VALUES (e.g. passing a string where an INT is expected)

Method 2: INSERT DATA USING SELECT — Moving Data From One Table to Another

→ *Copy data from the 'customers' table into 'persons'*

```
INSERT INTO persons (id, person_name, birth_date, phone)
```

```
SELECT
    id,
    first_name,
    NULL,
    'Unknown'
FROM customers
```

2. UPDATE — Modifying Existing Data

→ *Change the score of customer with ID 6 to 0*

```
UPDATE customers
SET score = 0
WHERE id = 6
```

→ *Update score and country for customer with ID 10*

```
UPDATE customers
SET score    = 0,
    country = 'UK'
WHERE id = 10
```

→ *Update all customers with a NULL score by setting their score to 0*

```
UPDATE customers
SET score = 0
WHERE score IS NULL
```

→ *Verify the update — should return no rows if successful*

```
SELECT *
FROM customers
WHERE score IS NULL
```

3. DELETE — Removing Data from Tables

→ *Preview customers with an ID greater than 5 before deleting*

```
SELECT *
FROM customers
WHERE id > 5
```

→ *Delete all customers with an ID greater than 5*

```
DELETE FROM customers
WHERE id > 5
```

→ *Delete all data from the persons table*

```
DELETE FROM persons
```

→ *Faster method to delete all rows (great for large tables)*

TRUNCATE TABLE persons